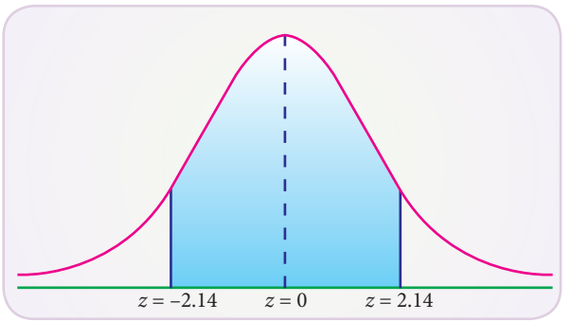


Fig 7.18

$P(3.5 < X < 6.5)$

$= P(-2.1429 < Z < 2.1429)$

$= P(0 < Z < 2.1429) + P(0 < Z < 2.1429)$

$= 2\ P(0 < Z < 2.1429)$

$= 2 \times .4838$

$= 0.9676$

Example 7.29

A sample of 125 dry battery cells tested to find the length of life produced the following resultd with mean 12 and SD 3 hours. Assuming that the data to be normal distributed , what percentage of battery cells are expected to have life

(i) more than 13 hours

(ii) less than 5 hours

(iii) between 9 and 14 hours

Solution :

Let X denote the length of life of dry battery cells follows normal distribution with mean 12 and SD 3 hours

(i) more than 13 hours

$P(X > 13)$

When $X = 13$

$$Z = \frac{X-\mu}{\sigma} = \frac{13-12}{3} = 0.333$$

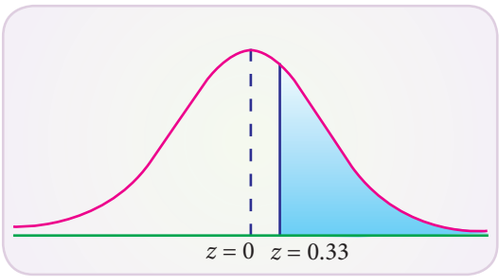


Fig 7.19

$P(X > 13) = P(Z > 0.333) = 0.5 - 0.1293 = 0.3707$

The expected battery cells life to have more than 13 hours is $125 \times 0.3707 = 46.34\%$

(ii) less than 5 hours

$P(X < 5)$

When $X = 5$ $Z = \frac{X-\mu}{\sigma} = \frac{5-12}{3} = -2.333$

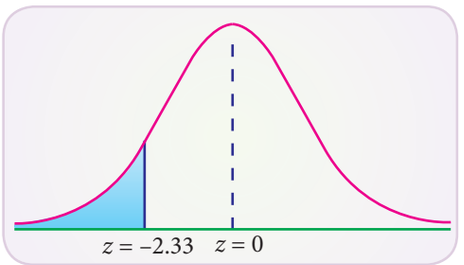


Fig 7.20

$P(X < 5) = P(Z < -2.333) = P(Z > 2.333)$

$= 0.5 - 0.4901 = 0.0099$

The expected battery cells life to have more than 13 hours is $125 \times 0.0099 = 1.23\%$

(iii) between 9 and 14 hours

When $X = 9$

$$Z = \frac{X-\mu}{\sigma} = \frac{9-12}{3} = -1$$

When $X = 14$

$$Z = \frac{X-\mu}{\sigma} = \frac{14-12}{3} = 0.667$$

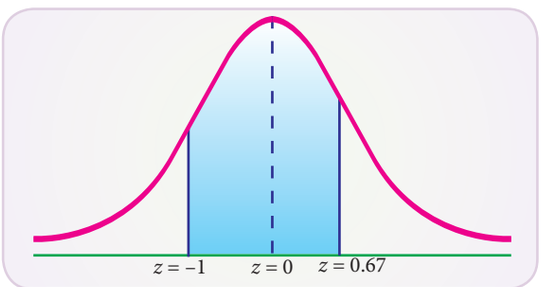


Fig 7.21

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